

KOMMUNAR, Russian Federation

WMO: 297590

Lat: 54.333N

Lon: 89.283E

Elev: 2778

StdP: 13.28

Time Zone: 7.00 (E07)

Period: 96-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-22.8	-17.9	-28.0	1.3	-22.6	-23.2	1.7	-17.6	16.7	6.7	14.5	8.3	1.5	270	0.621

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	16.3	78.5	62.2	75.0	61.0	71.9	59.4	64.6	74.4	62.8	72.1	61.1	69.4	2.5	90

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
60.9	88.7	68.5	59.2	83.2	66.5	57.6	78.5	65.2	31.3	74.1	29.8	72.2	28.5	69.4	75.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 12.8	10.0	8.6	-25.8	85.1	7.5	3.5	-31.2	87.5	-35.6	89.6	-39.8	91.5	-45.3	94.0		
(5)			-26.2	68.6	7.5	2.7	-31.6	70.5	-36.0	72.1	-40.2	73.6	-45.6	75.5		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	32.7	4.8	9.1	21.4	34.1	44.7	57.3	59.8	56.2	44.9	32.1	17.4	8.8		
	DBStd	21.56	11.96	11.59	10.67	10.10	9.20	7.33	5.46	6.67	8.23	9.55	12.02	11.24		
	HDD50	7184	1402	1144	888	478	219	24	2	19	196	557	979	1276		
	HDD65	11837	1867	1564	1353	926	631	244	176	283	604	1019	1429	1741		
	CDD50	858	0	0	0	3	54	241	305	210	42	3	0	0		
	CDD65	37	0	0	0	0	1	11	14	10	1	0	0	0		
	CDH74	398	0	0	0	0	14	147	136	89	12	0	0	0		
	CDH80	52	0	0	0	0	1	17	18	15	1	0	0	0		
	WSAvg	2.6	3.1	3.1	3.2	2.6	2.6	2.0	1.6	1.6	2.0	2.7	3.5	3.9		
	PrecAvg	32.4	1.6	1.4	2.2	2.7	2.7	3.0	3.3	3.2	2.6	3.2	3.0	2.6		
	PrecMax	41.3	5.2	3.7	5.9	5.7	6.0	5.7	8.3	6.2	4.6	6.9	6.0	6.3		
(15) Precipitation	PrecMin	23.8	0.0	0.0	0.6	0.6	0.6	1.0	1.7	0.8	0.6	0.8	1.7	0.4		
	PrecStd	4.7	1.2	0.9	1.4	1.3	1.5	1.1	1.5	1.1	1.0	1.4	1.1	1.5		
	DB	32.0	36.3	54.0	64.9	75.7	82.5	82.5	81.7	75.3	62.1	46.4	32.2			
	MCWB	29.9	30.3	40.4	47.3	55.9	62.6	65.0	64.3	57.3	46.9	37.6	29.0			
	2% DB	28.0	31.0	45.4	58.8	70.3	78.1	77.8	75.7	67.8	55.2	39.7	28.8			
(20) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	MCWB	25.5	26.8	35.7	44.7	53.8	61.1	63.4	62.3	54.0	45.1	35.5	26.4			
	5% DB	24.5	27.7	40.2	53.9	65.7	74.6	74.3	71.6	61.9	50.1	35.3	25.9			
	MCWB	22.1	24.5	32.9	42.0	50.4	60.0	62.1	60.3	51.5	42.4	32.2	23.5			
	10% DB	20.7	24.9	36.0	49.5	60.8	71.0	70.9	67.7	57.5	45.7	32.0	23.3			
	MCWB	18.5	22.1	30.9	40.0	48.6	58.5	60.7	58.6	50.0	40.2	29.8	21.1			
	0.4% WB	30.5	31.7	41.2	48.8	58.2	65.6	67.4	66.4	59.3	48.9	40.8	30.2			
	MCDB	32.3	34.9	52.7	62.0	72.8	77.4	77.6	77.4	70.6	57.7	44.1	31.5			
	2% WB	26.0	27.3	36.6	45.7	54.7	63.3	65.0	63.4	55.4	45.9	35.7	26.9			
(28) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	MCDB	27.5	30.4	43.7	56.2	66.8	73.9	74.4	73.0	64.6	53.2	39.6	28.6			
	5% WB	22.2	24.7	33.7	43.1	52.2	61.5	63.4	61.3	52.9	43.2	32.5	23.7			
	MCDB	24.7	27.6	39.3	51.8	62.1	71.8	72.2	69.5	60.0	49.5	35.0	25.6			
	10% WB	18.6	22.1	30.9	40.6	49.8	59.6	61.6	59.6	50.5	40.3	29.7	21.3			
	MCDB	20.6	24.8	36.0	48.3	58.8	68.6	69.2	66.7	56.5	45.4	32.1	23.3			
	MDBR	11.5	14.3	17.1	17.5	17.8	18.9	16.3	16.5	15.9	12.5	10.7	10.7			
	5% DB	13.0	15.3	23.2	24.6	26.8	26.1	23.1	23.6	24.9	20.2	12.3	12.1			
	MCWBR	11.2	12.0	15.5	13.8	13.3	12.2	10.7	11.3	13.0	12.2	9.2	10.6			
(36) Mean Daily Temperature Range	5% WB	13.0	14.0	21.7	22.8	22.3	22.0	19.7	20.6	21.4	18.7	11.7	11.8			
	MCWBR	11.4	11.4	15.2	13.5	12.1	11.0	10.2	10.8	12.0	12.0	9.2	10.7			
	taub	0.252	0.285	0.331	0.372	0.379	0.376	0.390	0.376	0.342	0.321	0.284	0.252			
	taud	2.115	2.087	2.024	2.134	2.257	2.362	2.333	2.349	2.397	2.311	2.118	2.044			
(42) Clear-Sky Solar Irradiance	Ebn at Noon	219	251	262	269	274	276	269	266	262	237	199	186			
	Edh at Noon	21	30	41	43	40	36	37	34	29	24	21	18			
	RadAvg	296	607	1034	1412	1611	1815	1617	1355	932	505	263	187			
(45) All-Sky Solar Radiation	RadStd	37	60	88	84	199	174	111	130	108	55	40	28			

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (14 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page